5 November 2026

Dear Editors,

Please consider "Quantum resources from localized quantum-field measurements: extraction and certification" as a Topical Review for *Frontiers of Physics*.

The review examines how a localized field interaction supports a resource claim in a specified retained probe. It connects causal measurement constructions, field-to-probe accessibility, entanglement harvesting, and calibrated readout. Recent Gaussian access results, interaction-induced resource attribution, and microscopic detector proposals motivate this focused synthesis for quantum-information and quantum-physics readers.

Existing measurement reviews explain the admissibility and construction of local operations; resource-theoretic work establishes interaction-dependent attribution conditions. This review follows the subsequent inference to a specified probe resource and experimental evidence. Its examples expose assumptions and limitations of selected diagnostics, including mixed-state residual negativities, perturbative revivals, and a generalized curvature–Fisher-information relation. It credits the underlying channel and statistical results and makes no claim to a new universal resource theory.

The technical proofs and numerical benchmarks remain in Appendices A–G within the same manuscript. The article contains five figures, seven tables, and 125 references. Reproducibility files provide input data, finite-state checks, independent quadrature reconstruction, and plotting code; no experimental data or uniform all-orders continuum error bound is claimed. An abstract graphic is supplied.

OpenAI ChatGPT/Codex (a GPT-6-based assistant provided by OpenAI, accessed November 2026) assisted literature organization, explanatory drafting, bilingual editing, citation checks, and Python/Matplotlib code refinement, for technical-writing and reproducibility support. This assistance and the computational checks are disclosed in the manuscript. The author remains responsible for the content and verification of AI-assisted material. No generative image model was used.

The review PDF and this letter omit author-identifying details for double-anonymized review; author information is supplied separately to the Editorial Office.

Thank you for considering the manuscript.

The author